

Priority Queue

Architecture of Real-Time Systems
SE 456

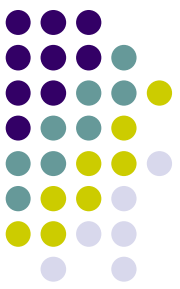
Ed Keenan

27 January 2021

13.0.8.3.19 Mayan Long Count

Animation System

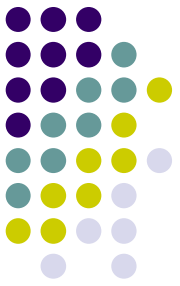
What's is it?



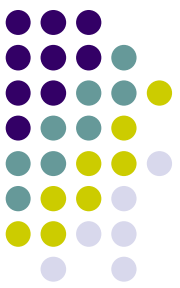
- Animation System
 - A systematic way to animate anything
 - Doesn't have to necessarily be aliens images
 - Anything that varies with time
 - Sequentially cycle between a grouping of images
 - Gives the allusion of animation
 - Timed based
 - Animations can trigger by an programmatic timer

Animation System

Architectural Design

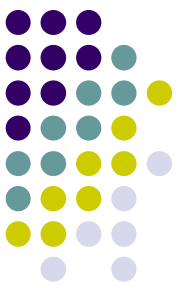


- Animation system breaks into a few parts
 - Timer
 - Need one timer to rule them all
 - Need singleton manager
 - Need update
 - Call time events when triggered
 - Priority Queue
 - Attached variable callbacks when timer triggers
 - Command Pattern
 - Sprite Animation object
 - Hold several images together
 - Deals with the adding to Timer system
 - Swaps image on a texture for a specified sprite

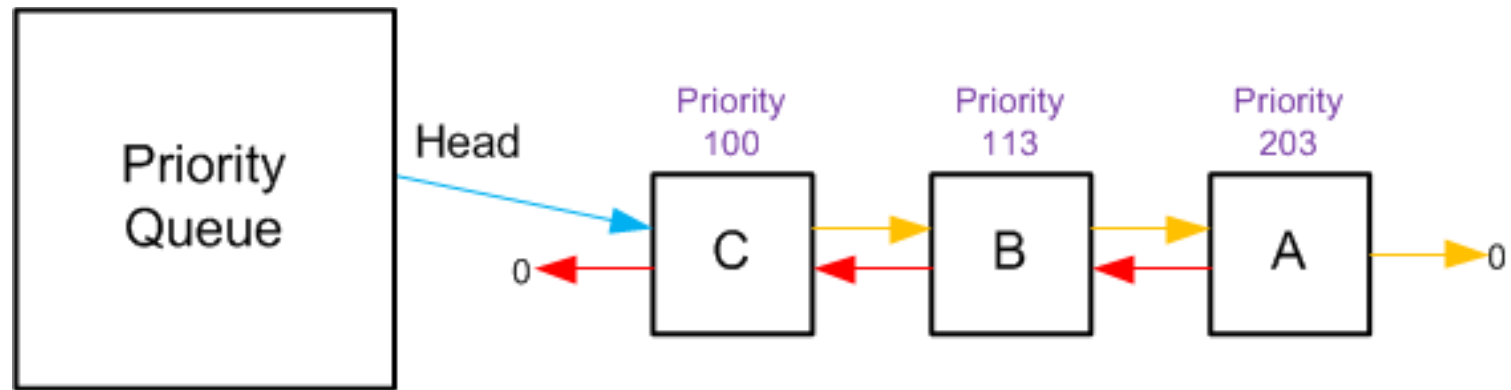


Priority Queue

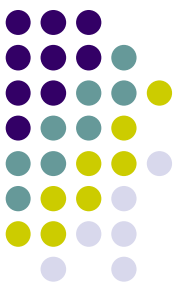
- A regular queue structure, but where additionally each element has a "priority" associated with it.
 - An element with high priority is served before an element with low priority.
 - If two elements have the same priority, they are served according to their order in the queue.
- Must at support:
 - `insert_with_priority`:
 - add an element to the queue with an associated priority
 - `pull_highest_priority_element`:
 - remove the element that has the highest priority and return it



Priority Queue



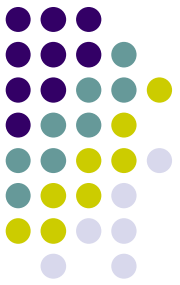
- Insert in priority order
- Execute the nodes based on priority
- Remove after execution



Priority Queue in Game Engine

- Where do you deal with Priority Queue?
 - Anything that can be triggered by a timer event
 - Animation Systems
 - Background events
 - Explosion duration
 - Complex movement
 - Have movement be triggered at a specific time
 - Sound effects
 - Random event scheduling
 - Schedule the UFO to play at a random time
 - Audio Squeaks by for basketball games\

Thank You!



- Questions?

